

The additional set of arguments available to the *xyplot* and *barchart* are listed below:

Argument	Description
<i>box.ratio</i>	Specifies the ratio of the width of rectangles in barchart
<i>panel</i>	Plots x and y variables in each panel
<i>default.prepanel</i>	A default function as a fallback to the prepanel function
<i>auto.key</i>	Used to produce a suitable legend
<i>aspect</i>	The physical aspect ratio of the panels
<i>axis</i>	A function responsible for drawing the axis annotation
<i>horizontal</i>	The orientation of the bar chart
<i>subscripts</i>	A logical flag to pass a 'subscripts' vector to the panel function
<i>subset</i>	A set of rows from the data is used in the plot

Scatter plot

You can also display individual charts on a panel grid. For example, the all India consumer price index (rural/urban) data set up to November 2021 is available from <https://data.gov.in/catalog/all-india-consumer-price-index-ruralurban-0> for the different states in India. We can read the data from the downloaded file using the *read.csv* function, as shown below:

```
> cpi <- read.csv(file="CPI.csv", sep=",")

> head(cpi)
  Sector Year      Name Andhra.Pradesh Arunachal.Pradesh
Assam Bihar
1   Rural 2011 January   104           NA      104      NA
2   Urban 2011 January   103           NA      103      NA
3 Rural+Urban 2011 January   103           NA      104      NA
4   Rural 2011 February  107           NA      105      NA
5   Urban 2011 February  106           NA      106      NA
6 Rural+Urban 2011 February  105           NA      105      NA
  Chattisgarh Delhi Goa Gujarat Haryana Himachal.Pradesh
Jharkhand Karnataka
1    105    NA 103   104    104    104    105    104
2    104    NA 103   104    104    103    104    104
3    104    NA 103   104    104    103    105    104
4    107    NA 105   106    106    05    107    106
```

```
5    106    NA 105   107    107    105    107    108
6    105    NA 104   105    106    104    106    106
```

The *aggregate* function can be used to obtain the values for the state of Andhra Pradesh as follows:

```
ap <- aggregate(x=cpi$Andhra.Pradesh, by=list(cpi$Year),
FUN=sum)
```

```
> head(ap)
  Group.1      x
1   2011 3911.28
2   2012 4255.40
3   2013 4516.60
4   2014 4673.60
5   2015 4822.20
6   2016 4921.50
```

A simple scatter plot can be displayed for the consumer price indexes using the following arguments to the *xyplot* function:

```
> xyplot(x~Group.1, ap, main="Andhra Pradesh Consumer Price
Index upto November 2021", xlab="Year", ylab="Consumer Price
Index")
```

The corresponding scatter plot illustration is shown in Figure 3.

Panel grid

You can also visualise the values per year (Group.1) using the *xyplot*:

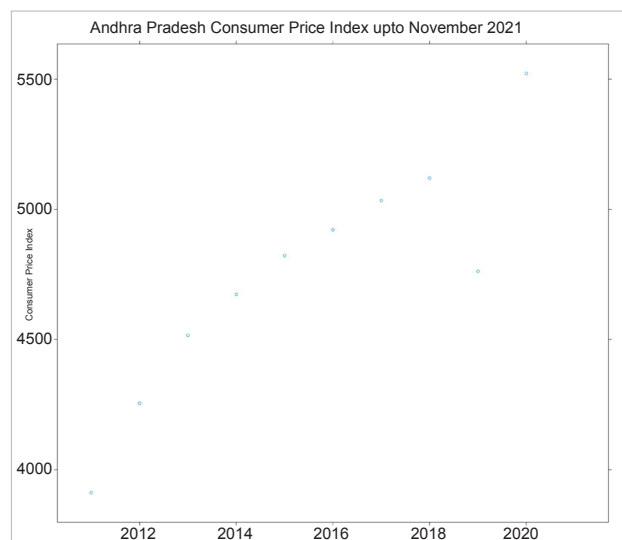


Figure 3: Scatter plot